United airlines flight 585

Flight model: Boeing 737
Date: March 03, 1991

From: Denver, US
To: Colorado, US

The Boeing 737 is considered the backbone of the aviation industry and is highly popular among buyers. There is heavy wind. The plane also experiences violent turbulence.

Problem

During the landing approach, as speed decreases the plane becomes more vulnerable to the turbulence. As they approach the airport the turbulence keeps increasing. suddenly the plane spins out of control. The plane crashes within 10 seconds.

Investigation

The investigators find the **PCU (Power control unit)** which is used to control the plane and act as a power steering. When the pilot apply gentle pressure in the rudder peddle, this PCU interprets this and moves the rudder to the required amount. The heart of the PCU, the valve called **dual servo valve** is used to direct the flow of pressure that moves the rudder. Technicians find chips of metal in the hydraulic fluid which can jam the dual servo valve. Both the PCU and dual servo motor are damaged due to crash. But, the makers of the dual servo value rules out the possibility sighting that a filter, filters out any foreign object in the fluid. The investigators were not able to determine the cause of the accident due to lack of enough evidence. But, years later a similar accident occurred US air flight 427 to Pittsburgh had experienced the same which made the plane to crash. In this crash they were able to recover the rudder hydraulic system and suffered little damage. In both the cases the rudder was deployed fully to one side which is called **rudder hardover**. Which again was left undermined with lack of evidence. Later, Eastwind Airlines Flight 517 had experienced the same. The pilot was not able to control the rudder as it was hard to push the rudder peddle but managed to control the yoke so that the plane does not roll over. This lasted 30 seconds and made the plane go to normal position. Then again flight 517 is pushed to the same side for 30 seconds the again goes back to normal then manages to land safely. Then few researchers who work on military plane found that when the dual servo valve is subjected to -40 degree temperature similar to temperature in the atmosphere when flying and when hot hydraulic fluid it passed the dual servo motor behaved differently. It stops for a while in full rise position just like it is jammed. This when tested at normal conditions work perfectly fine and would never face this problem. Also, one of the Boeing engineers found out that at these particular cases and condition the dual servo motor had not only jammed but also reversed. Which means when the pilots try to turn the rudder to the left after the experiences rudder hardover, instead of turning left the rudders turned further right.

Changes that were brought after the incident

New training was given to pilots to react to unusual events. And in the 737 training, the pilots were given training on rudder hardovers and rudder reversals. FAA also directed the Boeing to design the dual servo value to eliminate such rudder reversals.

Reference

1. Worst Equipment Failures in Aviation History — Mayday: Air Disaster (1:40:00 - 2:30:00)
   https://youtu.be/_aIR6BLdZ7Q